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The Million Dollar Hypothesis No One Can Prove

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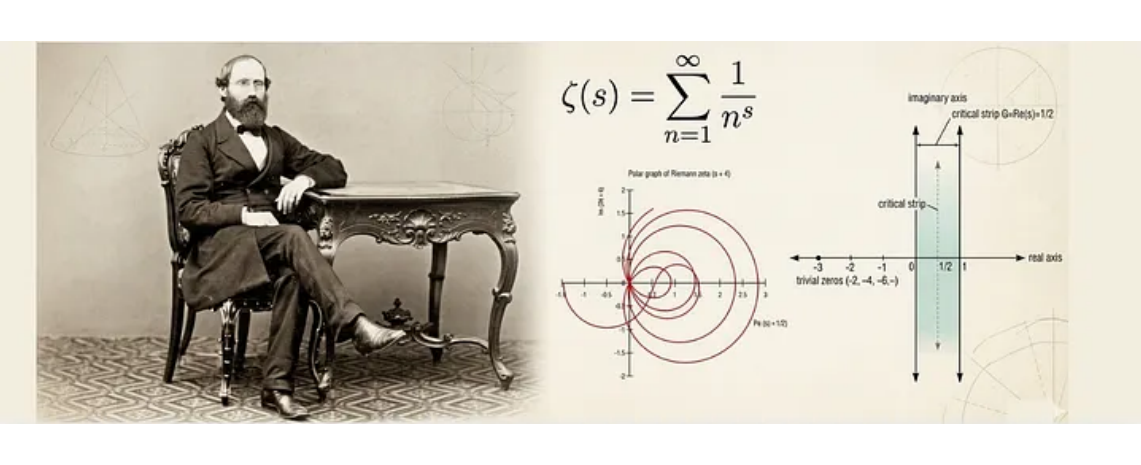
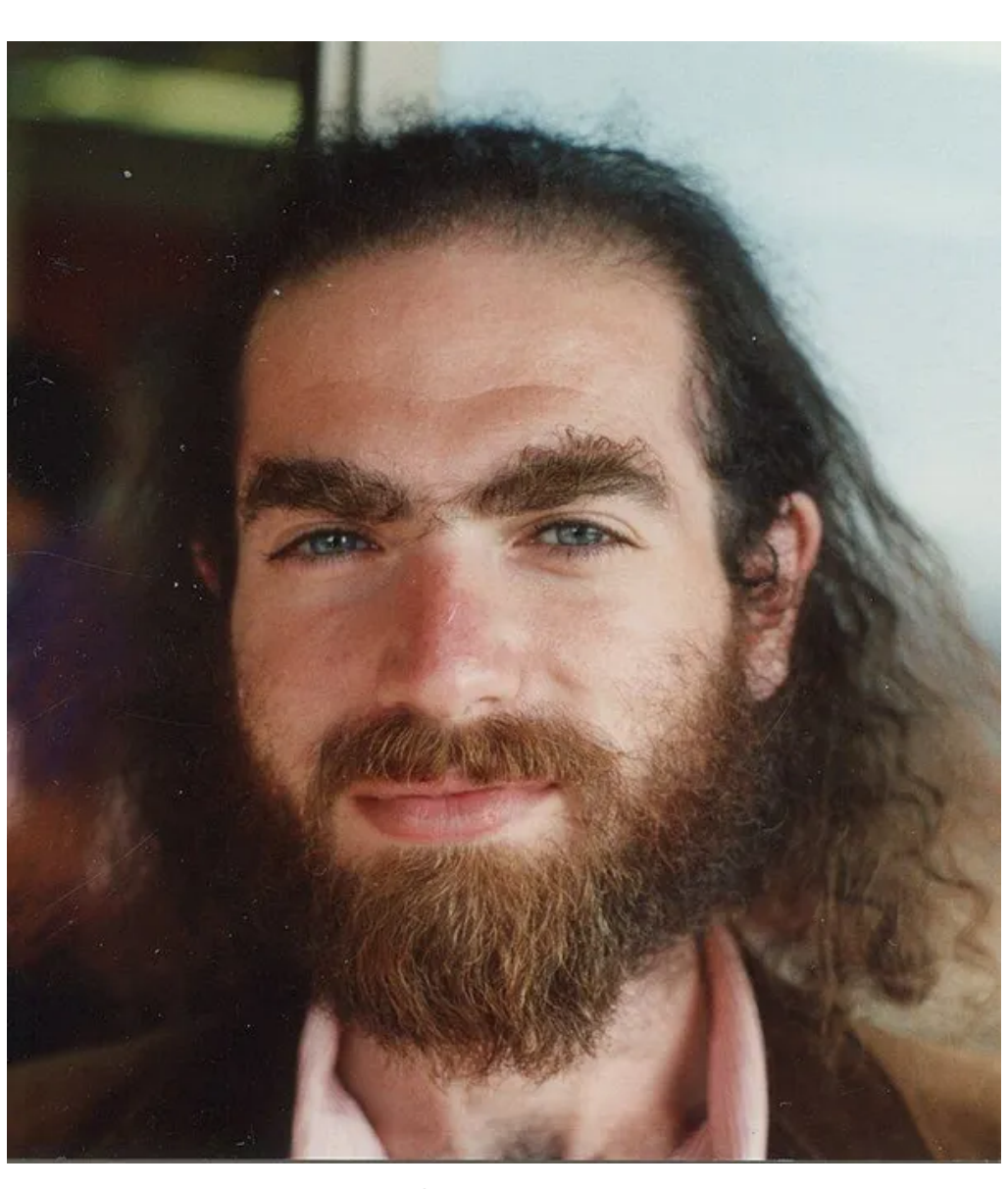
The Riemann Hypothesis stands as one of the most profound and alluring mysteries in human thought. To understand why a 167-year-old conjecture about a strange mathematical function has become the Mount Everest of modern mathematics, we need to build the concept piece by piece.


$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$
imaginary axis
critical strip
real axis
trivial zeros (-2, -4, -6,...)
1/2

A Prize Worth More Than Money

In the year 2000, the Clay Mathematics Institute announced seven of the hardest unsolved problems in mathematics — the "Millennium Prize Problems" — and offered one million dollars for the solution to each. Only one has been solved so far: the Poincaré Conjecture, cracked by the reclusive Russian mathematician Grigori Perelman in 2003.



Grigori Perelman

Remarkably, Perelman turned down both the million-dollar prize and the field's highest honor, the Fields Medal. His refusal says something important about how mathematicians actually think: the deepest problems are pursued not for money, but for the truth they promise to reveal.

Of the six problems that remain, one now stands as the single most famous open question in all of mathematics. That problem is the **Riemann Hypothesis**, and its story begins in 1859, with a short, dense paper written by the German mathematician Bernhard Riemann.

Bernhard Riemann (1826–1866). Source: Mondadori Portfolio / Mondadori via Getty Images

Part 1: From an Old Puzzle to a New Function

The Basel Problem

In the early 18th century, mathematicians were fascinated by a deceptively simple question. What do you get if you add up the reciprocals of all the square numbers, forever?

$$1+\frac{1}{4}+\frac{1}{9}+\frac{1}{16}+\frac{1}{25}+\cdots$$

Each term you add gets smaller, and the running total creeps closer and closer to some fixed number without ever running off to infinity. Mathematicians call this a convergent series — one whose partial sums approach a specific value, called the limit. Finding that limit stumped the best minds of the era for decades until the brilliant Swiss mathematician Leonhard Euler cracked it. His answer stunned everyone:

$$1+\frac{1}{4}+\frac{1}{9}+\frac{1}{16}+\cdots=\frac{\pi^2}{6}$$

Why should π — a number defined by circles — appear in a sum built entirely from whole numbers? Euler's proof was ingenious, and it hinted that something deep connected the world of counting numbers to the world of geometry.

A Function That Generalizes the Pattern

Euler's sum is really just one instance of a more general pattern. Instead of always squaring the denominators, what if we raised them to any power s? This gives us the Riemann zeta function, written ζ(s):

$$\zeta(s)=\frac{1}{1^s}+\frac{1}{2^s}+\frac{1}{3^s}+\frac{1}{4^s}+\cdots$$

For any value of s greater than 1, this sum converges to some definite number. Set s = 2, and you recover Euler's $\pi^2/6$. The zeta function is effectively a machine: feed it a number s, and it hands back another number.

Part 2: Into the Complex Plane

What Happens When the Series Breaks?

Curiosity naturally leads to a dangerous question: what if we plug in s = −1? The formula appears to produce:

$$1+2+3+4+\cdots$$

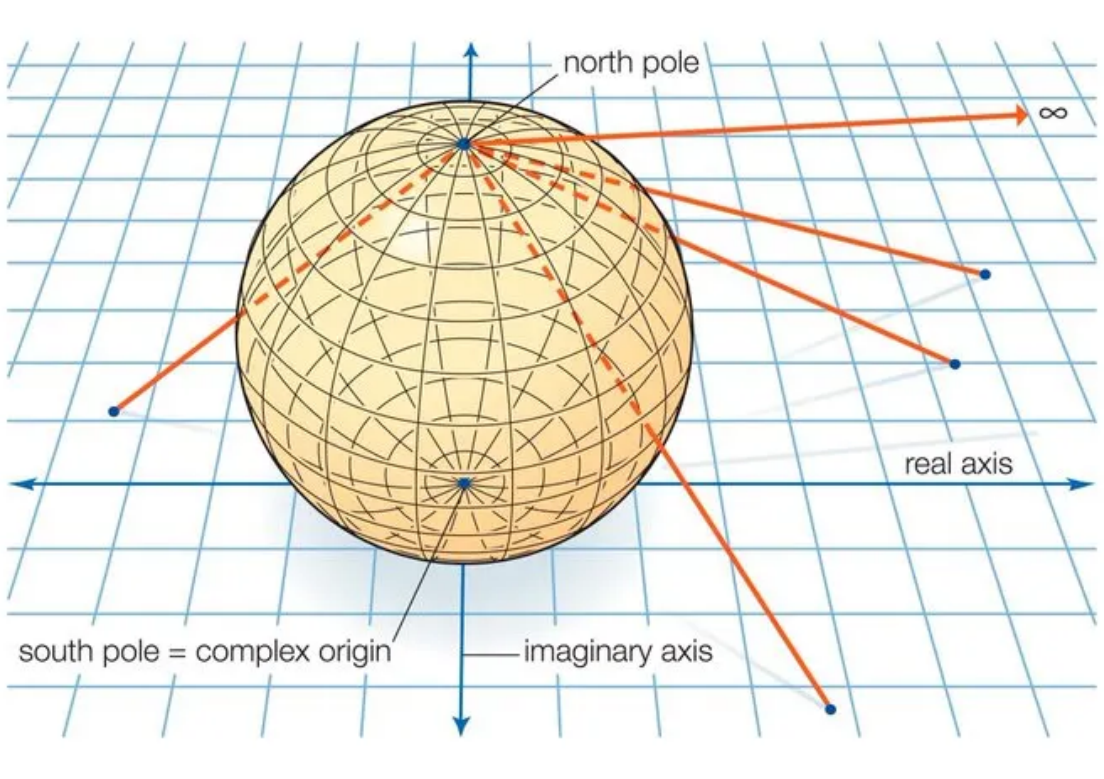
This is the famous "sum of all natural numbers." It is a divergent series, meaning the partial sums grow without bound, never settling near any fixed number. On its face, there's no sensible way to assign this sum a finite value.

Yet, astonishingly, when this expression is understood in the context of the zeta function rather than as a naive sum, mathematics does provide a way to assign it a specific, finite value. That value turns out to be −1/12 — but to see how, we first need to expand the universe of numbers s is allowed to live in.

Enter Complex Numbers

Ordinary "real" numbers can be pictured as points on a single line: 0, 1, 2, fractions like ½, and π all sit somewhere on that line. But there is no point on this line representing √−1, because no real number multiplied by itself gives a negative result.

Rather than banning √−1, mathematicians long ago found a more powerful solution: give it its own dimension. By drawing a second axis perpendicular to the real number line, we create the complex plane.



Visualizing the Complex Plane. Source: Encyclopaedia Britannica / Universal Images Group via Getty Images

Every point in this resulting plane represents a number with two parts: a real part (horizontal) and an imaginary part (vertical). This might sound like abstract bookkeeping, but it is precisely the tool Riemann needed.

Part 3: Riemann's Great Leap

Extending the Function Into the Plane

Riemann's insight was to stop treating s as merely a whole number or even a real number, and instead let it range over the entire complex plane. Within the region where the real part of s is greater than 1, ζ(s) behaves as a beautifully smooth and well-mannered function. Functions with this property enjoy a remarkable trick called analytic continuation: it's possible to extend their definition outward, beyond the region where the original formula mathematically converges, in a way that is completely unique and unambiguous.

In his 1859 paper, Riemann carried out exactly this extension. He showed that ζ(s) can be defined for *every* complex number — with a single, solitary exception at s = 1, where the function hits an infinite roadblock called a pole.

Once this extension is in place, we can finally return to that puzzle from before. Plugging the extended zeta function's rules at s = −1 does indeed yield a single, well-defined, computable answer:

$$\zeta(-1)=-\frac{1}{12}$$

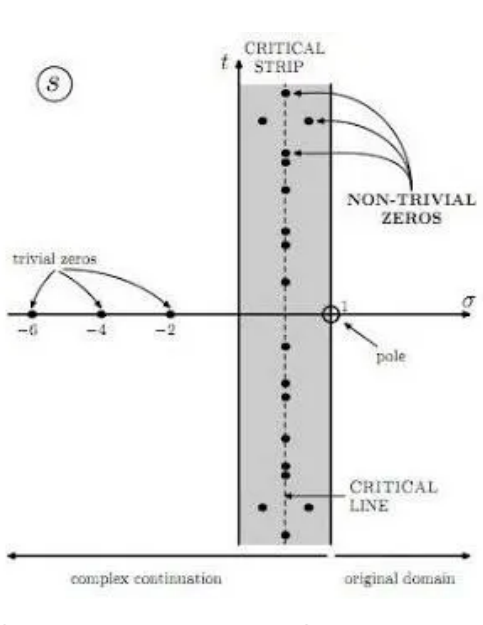
This is the rigorous sense in which mathematicians say "1 + 2 + 3 + 4 + ⋯ = −1/12" — not as a literal sum of blocks, but as the value the mathematically extended zeta function takes at that exact coordinate.

The Zeros — and the Hypothesis Itself

With ζ(s) now defined across the whole complex plane, a natural question arises: for which values of s does ζ(s) equal exactly zero?

Some zeros are easy to find — the function vanishes at every negative even integer: −2, −4, −6, and so on. Mathematicians call these the "trivial zeros." The far more interesting zeros are the ones that don't fit this pattern. It has been proven that every such "non-trivial" zero must lie within a specific band of the complex plane called the critical strip — consisting of all complex numbers whose real part is strictly between 0 and 1.

Running straight down the center of this strip is the critical line, where the real part equals exactly ½.



The Critical Line at Real Part = ½. Source: ResearchGate

> **The Riemann Hypothesis states:** Every non-trivial zero of the zeta function lies exactly on the critical line.

That's the entire statement. It is deceptively simple, yet unproven since 1859. To disprove it, someone would only need to locate a single non-trivial zero anywhere in the critical strip that wanders off that central line. Supercomputers have verified that the first trillions of zeros do indeed sit perfectly on the critical line, with not a single counterexample found — powerful evidence, but not a mathematical proof.

Why It Matters: Primes in Disguise

The Riemann Hypothesis might sound like an abstract curiosity confined to the world of complex graphs, but its true significance lies elsewhere entirely: in the distribution of prime numbers.

Primes — numbers divisible only by 1 and themselves like 2, 3, 5, 7, 11 — have fascinated mathematicians for over two thousand years. Their pattern among the integers appears maddeningly irregular, like weeds scattered randomly in a field.

In that same 1859 paper, Riemann discovered a stunning bridge between his zeta function and the primes. He derived a formula expressing precisely how many primes exist below any given number n built directly from the zeros of ζ(s).

> It is a genuine surprise that a question rooted in the geometry of the complex plane should govern something as concrete as counting prime numbers, a subject belonging to an entirely different branch of mathematics.

Crucially, the accuracy and full power of Riemann's prime-counting formula depend on the Riemann Hypothesis being true. If all non-trivial zeros really do lie perfectly on the critical line, the formula's error margins shrink to their smallest possible size, giving mathematicians the most precise understanding achievable of how primes are scattered among the integers.

This is what elevates the Riemann Hypothesis from an elegant puzzle to arguably the central open question in all of mathematics — a hidden thread connecting two seemingly unrelated worlds, still waiting, over a century and a half later, to be pulled all the way through.

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What are your thoughts?

Greg Herlihy
3 days ago

I have discovered a truly marvelous proof of this Riemann Hypothesis, which this comment text box is too small to contain.

9 1 reply Reply

Sanjay Singh
1 day ago

> That value turns out to be −1/12

Ramanujan has done work with infinite series where this number also appears, right?

1 Reply

Sanjay Singh
1 day ago

> Remarkably, Perelman turned down both the million-dollar prize and the field's highest honor, the Fields Medal.

There are both personal and professional reasons for why he did this. Please read the New Yorker, what I believe is the definitive account of what happened. But we warned, some of it is ugly:

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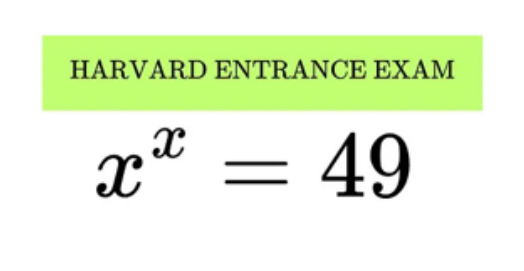

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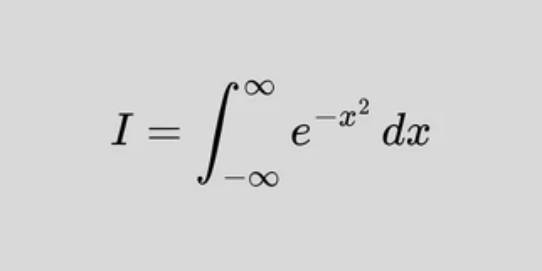

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